

What is Interest

IF $\$K$ grows to $\$S$ then $\$S - \K is interest

"rent paid by borrower for the use of $\$K$ "

"compensation to a lender for the loss of choice"

Money changes as time changes so lets call $t=0$ the time of investment.

Def. The accumulation of $\$K$ until time t is defined as $A_K(t) \leftarrow \text{domain } \{t: t \geq 0\}$

$A_K(t)$ is called the amount function.

IF $K=1$ we write $a(t) := A_1(t)$. $a(t)$ is called the accumulation function.

Typically:

$$A_K(t) = K a(t)$$

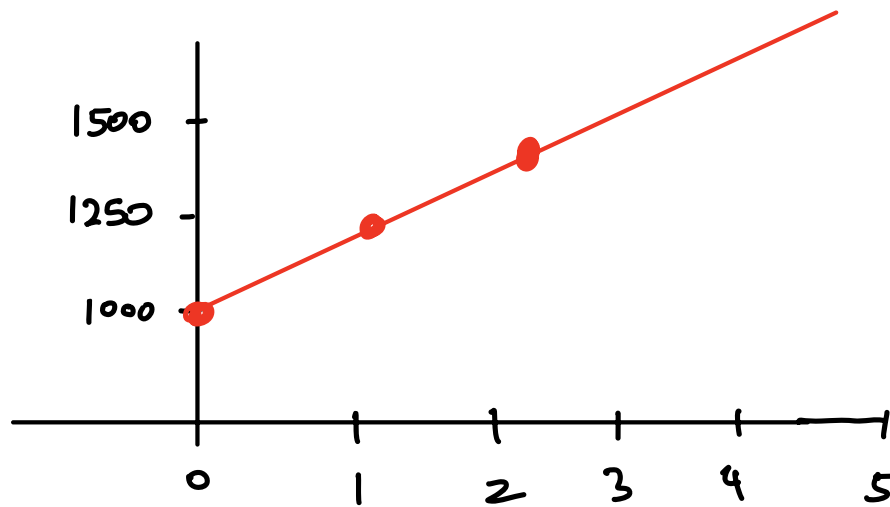
unless otherwise stated!

The following are always values of $A_K(t)$ and $a(t)$

- $A_K(0) = K$

- $a(0) = 1$

Example \$1000 grows by constant amount of \$250 each year for 5 years. Graph $A_{1000}(t)$



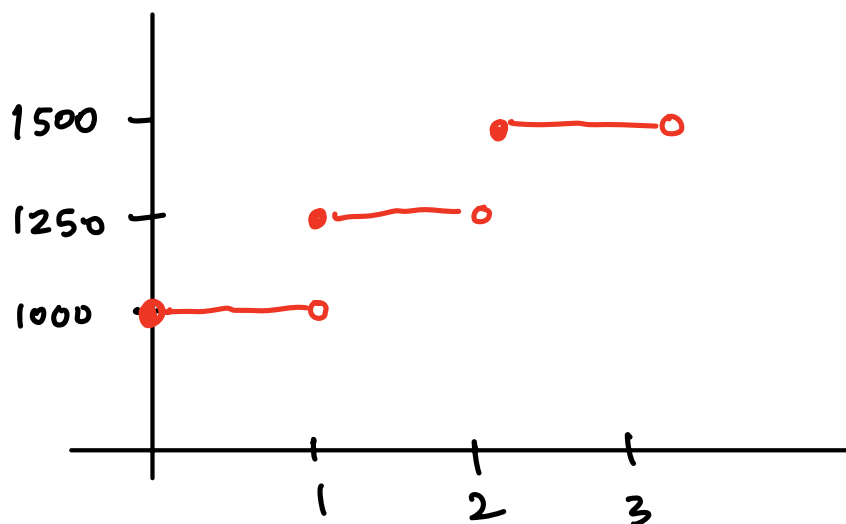
Slope: $\frac{\Delta y}{\Delta x} = 250$ $A_{1000}(t) = 1000 + 250t$

$a(t) = 1 + \frac{250}{1000}t$

↑
amount

↑
rate

what happens if interest is paid at EOY?



By the previous example, $A_{1000}(t)$ encodes info for the amount accumulated whereas $a(t)$ tells us about rates.

Def. Given times $0 \leq t_1 < t_2$

- Amount of interest: $A_k(t_2) - A_k(t_1)$
- effective interest rate:
$$\frac{a(t_2) - a(t_1)}{a(t_1)}$$
$$= \frac{A_k(t_2) - A_k(t_1)}{A_k(t_1)}$$

When we have n time periods $1, 2, \dots, n$

$[n-1, n] := n^{\text{th}}$ time period

$$i_n := \frac{a(n) - a(n-1)}{a(n-1)}$$

$$i_1 := \frac{a(1) - a(0)}{a(0)} = a(1) - 1$$

1.4 Simple Interest (Linear)

let's think about a linear accumulation function

$$a(t) = mt + b$$

1. what's b ? $a(0) = 1 \Rightarrow b = 1$

$$a(t) = 1 + mt$$

2. what's m ?

well... $a(1) = 1 + m \Rightarrow m = \underbrace{a(1) - 1}_{\substack{|| \\ i_1}}$

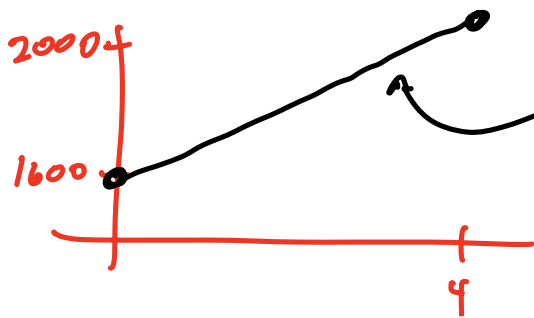
so all simple Interest accumulation functions:

$$a(t) = 1 + \overset{\text{interest rate!}}{i_1} t$$

$$A_K(t) = K(1 + i_1 t)$$

Problem: Alice lends Bob \$1600. Bob will repay \$2000 after 4 years according to simple interest. What is the interest rate?

Solution.



$A_{1600}(t)$

$$A_{1600}(t) = 1600 + mt$$

$$m = \frac{\Delta y}{\Delta x} = \frac{400}{4} = 100$$

$$A_{1600}(t) = 1600 + 100t$$

what's the skeleton?

$$1600 \left(1 + \frac{1}{16}t\right) \quad \text{so rate} = \frac{1}{16} = .0625$$

Problem: Alice loans Bob \$2400 for 3 years at 5% simple interest but Bob wants to repay loan early. Find the amount Bob needs to pay if he wants to end the loan after 3 years, 2 years, 1 year? If the loan lasts 3 years find i_1, i_2, i_3 .

Solution.



$$A_{2400}(t) = 2400(1 + .05t)$$

$$A_{2400}(3) = 2760$$

$$A_{2400}(2) = 2640$$

$$A_{2400}(1) = 2520$$

$$i_1 = \frac{A(1) - A(0)}{A(0)} = \frac{2520 - 2400}{2400} = .05$$

$$i_2 = \frac{A(2) - A(1)}{A(1)} = \frac{2640 - 2520}{2520} \approx .0476$$

$$i_3 = \frac{A(3) - A(2)}{A(2)} = \frac{2760 - 2640}{2640} \approx .04545$$

Given a simple interest $A_K(t) = K(1 + rt)$
what is i_n ?

$$i_n = \frac{A(n) - A(n-1)}{A(n-1)} = \frac{1 + rn - (1 + r(n-1))}{1 + r(n-1)}$$

$$i_n = \frac{r}{1 + r(n-1)}$$

decreasing sequence!
(Not as common in real world)

It is naive to think loans always come in yearly time steps. Here are some common practices where loans are time-stepped in days:

(1) Exact :
(Actual/Actual)

Take # of days in loan
days in a year

(2) Ordinary :
(30/360)

Each month = 30 days
 $\Rightarrow$ 360 days in a year

(3) Bankers Rule :
(Actual/360)

counts actual days but
imagines 360 days in
a year.

problem. Brad borrows 5,000 \$ from Julio on October 14th 1998 at 8% exact simple interest and repays the loan on May 7th 1999. How much does Brad pay?

Solution: # days

Months: 10, 11, 12, 1, 2, 3, 4, 5

2 partial
6 full

8

Oct: 31-14 Nov: 30 Dec: 31

Jan: 31 feb: 28 mar: 31 April: 30

may: 7

$$= 17 + 30 + 31 + 28 + 31 + 30 + 7$$
$$= 205$$

$$\$5000 \left(1 + .08 \left(\frac{255}{365} \right) \right) = \$5224.66$$

problem What if the previous problem was ordinary simple interest?

Oct: 30-14 Nov - April: 30 May: 7

$$16 + 7 + 30(6) = 180 + 23 = 203$$

$$5000 \left(1 + .08 \left(\frac{203}{360} \right) \right) \approx 5225.56 \$$$

problem What if the previous problem used the Banker's Rule?

$$5000 \left(1 + .08 \left(\frac{205}{360} \right) \right) \approx 5227.78$$

1.5 Compound Interest (Exponential)

As seen in the previous chapter, the effective interest rate in for simple interest is a decreasing sequence. So someone who invests at a bank would be incentivized to close an account just to instantly reopen a new one.

We need an accumulation s.t.

$\bar{i}_n$ doesn't depend on n

definition. $i = i_1 = a(1) - 1$

claim. If $i_n = i$ for all n then

$$a(t) = (1 + i)^t \quad \text{for all } t \in \mathbb{N}.$$

pf: $a(0) = 1 = (1+i)^0$ ✓

assume true up to k .

$$a(k+1) = (1+i)a(k) = (1+i)(1+i)^k = (1+i)^{k+1}$$

problem Bob deposits 12,000\$ into an account according to compound interest accumulation $a(t) = (1.05)^t$ he closes the account in 6.5 years How much is in it?

solution. $12,000(1.05)^{6.5} = 16,478.27\$$

Variations of Comp. Interest: $\lfloor x \rfloor = \text{round down.}$

①

problem: Bob deposits 12,000\$

$$a(t) = (1.05)^{\lfloor t \rfloor} (1 + .05(t - \lfloor t \rfloor))$$

describe in words the accumulation function and find 6.5 years again

solution Full years: compound interest
fractional years: simple interest

$$12000a(6.5) = 16,483.18$$

② Tiered Investment

problem.

$$i = \begin{cases} 2\% & \text{Account} < 2000\$ \\ 3\% & 2000 \leq \text{Account} < 5000 \\ 4\% & \text{Account} \geq 5000\$ \end{cases}$$

determine $A_{1800}(t)$ and

show $A_{1800}(t) \neq 1800a(t)$ even
though they agree for any t in $[0, 5.32]$

solution

to find $A_{1800}(t)$ we need to know when
the account exceeds \$2000 and when it
exceeds \$5000.

$$1800(1.02)^t = 2000$$

$$1.02^t = \frac{10}{9}$$

$$t = \ln\left(\frac{10}{9}\right) / \ln(1.02)$$

$$= 5.32 \text{ years}$$

once it's 2,000

$$2000(1.03)^t = 5000$$

$$t = \ln\left(\frac{5}{2}\right) / \ln(1.03)$$

$$= 30.9989 \text{ years.}$$

so

$$A_{1800}(t) = \begin{cases} 1800(1.02)^t & 0 \leq t \leq 5.32 \\ 2000(1.03)^{t-5.32} & 5.32 \leq t \leq 36.52 \\ 5000(1.04)^{t-36.52} & t \geq 36.52 \end{cases}$$

whereas $1800 a(t)$ is 1800 times

$$a(t) = \begin{cases} 1(1.02)^t & 0 \leq t \leq 383.833 \\ 2000(1.02)^{t-383.833} & 383.833 \leq t \leq 414.83 \\ 5000(1.02)^{t-414} & t \geq 414.83 \end{cases}$$

$$A_{1800}(t) \geq 1800 a(t)$$

1.6 Interest in advance / Discount Rates

Remember interest is like the rent we pay for borrowing money. When renting an apt. you pay rent before "using" the apt. Same w/ money.

def: If an investor lends $\$K$ for one basic period at a discount rate D then the borrower will have to pay $KD\$$ in order to receive the use of $\$K$

- $\$K$ $\leftarrow$ loan amount
- D $\leftarrow$ discount rate
- $\$KD$ $\leftarrow$ amount of discount
- $\$K - KD$ $\leftarrow$ amount borrower can use
 $= K(1 - D)$

Example Chad borrows \$1,000 at a discount rate of 7%. How much extra money does he have use of

solution: $1000 - 1000(.07) = 1000(.93) = 930\$$

def: Given a time interval $[t_1, t_2]$ the discount rate in the interval is

$$d_{[t_1, t_2]} = \frac{a(t_2) - a(t_1)}{a(t_2)}$$

In previous example: $7\% = \frac{1000 - 930}{1000}$

def: n is a positive integer $[n-1, n]$ is the n^{th} time period

$$d_n = \frac{a(n) - a(n-1)}{a(n)} \Rightarrow a(n-1) = a(n)(1 - d_n)$$

$\pm$

problem: Suppose $a(t) = (1.05)^2(1 + .025t)$
find d_4 and i_4

solution: $d_4 = \frac{a(4) - a(3)}{a(4)} = .0463$

$$i_4 = \frac{a(4) - a(3)}{a(3)} = .0485$$

clearly i_n and d_n are closely related.

def: a rate of interest and a rate of discount are equivalent for $[t_1, t_2]$ if for each \$1 invested, the two rates produce the same value at t_2

Consider a loan $\$L$ for $[t_1, t_2]$

at t_2 we must repay: $\$L(1 + i_{[t_1, t_2]})$

so in the beginning of loan the borrower walks away with

$$\$L(1+i_{[t_1, t_2]})(1-d_{[t_1, t_2]})$$

so $i_{[t_1, t_2]}$ is equivalent to $d_{[t_1, t_2]}$ if

$$\$L = \$L(1+i_{[t_1, t_2]})(1-d_{[t_1, t_2]})$$

$$\Rightarrow 1 = 1 + i_{[t_1, t_2]} - d_{[t_1, t_2]} - i_{[t_1, t_2]}d_{[t_1, t_2]}$$

$$\Rightarrow d_{[t_1, t_2]} = i_{[t_1, t_2]}(1 - d_{[t_1, t_2]})$$

so:

$$i_{[t_1, t_2]} = \frac{d_{[t_1, t_2]}}{1 - d_{[t_1, t_2]}} \quad d_{[t_1, t_2]} = \frac{i_{[t_1, t_2]}}{1 + i_{[t_1, t_2]}}$$

$$i_n = \frac{d_n}{1 - d_n}$$

$$d_n = \frac{i_n}{1 + i_n}$$

problem. Suppose account has compound interest at annual effective rate of 8%. Find an expression for d_n .

solution:
$$d_n = \frac{i_n}{1 + i_n} = \frac{.08}{1 + .08} = .074 = \frac{2}{27}$$

1.7 Discount Functions

Money today is worth more than money later

Ex: $\$K$ today $\rightarrow \$A_K(t)$ in t years
 $\$1$ today $\rightarrow \$a(t)$ in t years
 $\$ \frac{1}{a(t)}$ today $\rightarrow \$1$ in t years

definition We define the discount function

$$V(t) = \frac{1}{a(t)}$$

So $V(t_0)$ is the amount of money you must invest at time 0 to have \$1 at time t_0

Note: $V = \frac{1}{1+i}$

Problem. Suppose we have simple interest at 5% for five years. How much should you invest now to have \$23,000 three years from now?

Solution.

$$K(1 + .05(3)) = 23,000$$

$$K = 20,000$$

$$\text{or } v(3) = \frac{1}{a(3)} = \frac{1}{1 + .05(3)}$$

↑
How much to
invest for \$1

$$\text{so } 23,000 \left(\frac{1}{1 + .05(3)} \right)$$

↑
How much to invest for
\$23,000

Counter Intuitive! ↘

problem. Again suppose $a(t) = 1 + .05t$
If you wish to have 23,000
in 5 years how much to
invest in 2 years?

Solution

If we wanted 23,000 after 5 years:

$$K(1 + .05(5)) = 23000 \Rightarrow K = 18460$$

and 18460 grows to

$$18460(1 + .05(2)) = 20,240 \text{ in 2 years}$$

so we should invest 20,240 in 2 yrs to get
23,000 in 5 years.

Moral of the story:

The accumulation of money not only
depends on the length of the time
interval but also on where the time
interval lies!

Fact: How much to invest in t_1 years
to have $\$S$ in t_2 years

$\frac{S}{a(t_2)}$ $\leftarrow$ how much you'd need to
invest now.

$\frac{S}{a(t_2)} \cdot a(t_1)$ $\leftarrow$ how much that
initial investment is
in t_1 years.

$$= S \cdot a(t_1) \cdot v(t_2) \quad \text{or}$$

$$S \cdot \frac{v(t_2)}{v(t_1)} \quad \text{or}$$

$$S \cdot \frac{a(t_1)}{a(t_2)}$$

1.9 Compound Discount

In section 1.5 we considered when i_n is independent of n . This forces

$$a(t) = (1+i)^t$$

what if d_n is also constant (indep of n)

If $d_n = d$ for all n recall

$$i_n = \frac{d_n}{1 - d_n} = \boxed{\frac{d}{1 - d} = i}$$

equivalently

$$\boxed{i = \frac{1}{1 - d} - 1}$$

we call d the "effective discount rate"
for the basic time period

when the time period is one year

d = annual effective discount rate

problem. Given an annual effective discount rate of 4.386286% compute annual effective interest rate.

solution.

$$i = \frac{1}{1-d} - 1$$

$$= \frac{1}{.95613714} - 1 = 4.5875072\%$$

Note by solving the above equation for d we get

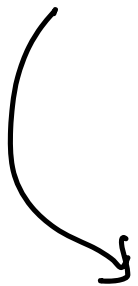
$$d = \frac{i}{1+i}$$

or

$$d = 1 - \frac{1}{1+i}$$

Equivalent Formulas:

$$d = i v, \quad d = 1 - v, \quad v = 1 - d$$



The amount of discount d on \$1 is equal to the amount of interest on \$1 discounted for one year by the discount factor.

problem. Which rate would you choose to borrow for your tuition?

(i) effective interest rate of 5.1%

(ii) effective discount rate of 4.9%

solution. (i) $i = 5.1\%$ $d = \frac{i}{1+i} = 4.8525\ldots\%$

(ii) $d = 4.9\%$ $i = \frac{d}{1-d} = 5.15247\ldots\%$

Choose (i) !!!

$$\underline{\underline{a(t) = (1-d)^{-t}}}$$

compound discount
accumulation function

(equal to $a(t) = (1+i)^t$)

problem.

EXAMPLE 1.9.13

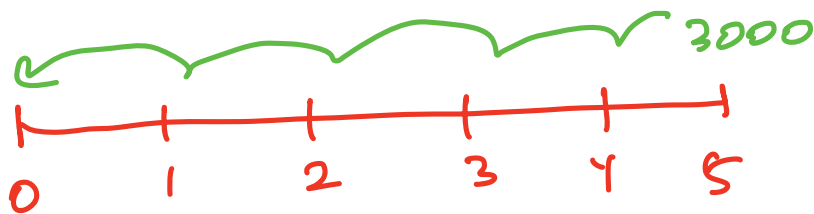
Problem: Ezra has the opportunity to borrow money according to compound discount at an annual effective discount rate of 8%. He would like to borrow money in order to completely pay for a \$3,000 used piano. Ezra wishes to repay the loan with a payment in exactly five years. How much money must he ask to borrow at 8% discount?

Solution.

$$d = .08 \Rightarrow i = \frac{.08}{.92} = 8.695652174$$

whoops this is rounded!!!
be careful!

$$\$3000 = X(1+i)^{-5} \Rightarrow X = \frac{3000}{(1 + \frac{.08}{.92})^{-5}}$$



borrower
perspective.

need to pay "rent" to borrow

$$= 4551.79 \$$$

$$\text{or } 3000(1-d)^5 \\ = 3000(.92)^5$$

1.10 Nominal Rates of interest/discount

Suppose we have compound interest. Then

$$a(t) = (1+i)^t = \underbrace{1}_{\text{initial value}} + \underbrace{[(1+i)^t - 1]}_{\text{interest}}$$

more specifically the effective interest rate for a period of length t

If the length of my period is $\frac{1}{m}$ we have

$$(1+i)^{\frac{1}{m}} - 1 \quad \text{as the effective interest rate.}$$

Banks typically credit interest more than once a year. They use a

Nominal Interest rate $i^{(m)}$ convertible
(or Compounded or payable m times per year)

Definition If an account is governed by a nominal interest rate of $i^{(m)}$ payable m times per year, the bank pays interest at a rate of

$$\frac{i^{(m)}}{m} = \left[(1+i)^{\frac{1}{m}} - 1 \right] \text{ per } m^{\text{th}} \text{ of a year.}$$

$$i = \left(1 + \frac{i^{(m)}}{m} \right)^m - 1$$

Banks report i as the Annual Percentage Yield or APY.

Since interest earned begins also making interest, $i^{(m)} < i$ for any $m > 1$.

EXAMPLE 1.10.4

Problem: National Bank advertises a savings account paying 4% nominal interest compounded quarterly. What is the annual effective yield **APY** for this account?

Solution.

$$\left(1 + \frac{.04}{4}\right)^4 - 1 = .04060401$$

Problem: Sandra inherits \$10,000. She deposits it in a five-year certificate of deposit paying 6% nominal interest compounded monthly and the interest remains on deposit. At the end of the five years, Sandra decides to renew her CD for another five years at the then current nominal interest rate of 7.5% compounded quarterly. Again, interest is left to accrue. At the time her second CD matures, what is her investment worth?

Solution.

6% monthly → $i = \left(1 + \frac{.06}{12}\right)^{12} - 1$
5 years $= 6.16778118645$

7.5% quarterly $i = \left(1 + \frac{.075}{4}\right)^4 - 1$
=

$$A(10) = 10,000 \left(1 + \frac{.06}{12}\right)^{60} \left(1 + \frac{.075}{4}\right)^{20} = 19557.626$$

Note if bank rounded after 1st CD then

$$A(10) = \underbrace{10,000 \left(1 + \frac{.06}{12}\right)^{60}}_{\text{Rounded}} \rightarrow \times \left(1 + \frac{0.075}{4}\right)^{20}$$

1 penny less!!!

Problem: Vladimir deposits \$50,000 in a three-year certificate of deposit for which interest is compounded quarterly and is left to accrue. At the end of the three years, the balance in the CD is \$63,786.11. What is the nominal annual interest rate $i^{(4)}$ convertible quarterly?

Solution. $63,786.11 = 50000 \left(1 + \frac{i^{(4)}}{4}\right)^{12}$

$$i^{(4)} = 4 \left(\left(\frac{63786.11}{50000} \right)^{\frac{1}{12}} - 1 \right) \approx 8.2\%$$

Problem: Jolene invests money in a fund for which interest is paid once every two years. The effective rate per two-year period is 14%. Find the nominal interest rate convertible biennially⁷ and the annual effective interest rate governing the fund.

$$\left(1 + \underbrace{\frac{i^{(1/2)}}{1/2}}_{= 14\%}\right)^{\frac{1}{2}} = 1 + i$$

$$i^{(1/2)} = 7\%$$

$$i = 6.77\%$$

Definition If an account is governed by a nominal **Discount** rate of $d^{(m)}$ payable m times per year, the bank pays **discount** at a rate of

$$\frac{d^{(m)}}{m} = \left[1 - (1 - d)^{\frac{1}{m}} \right] \text{ per } m^{\text{th}} \text{ of a year.}$$

$$d = 1 - \left(1 - \frac{d^{(m)}}{m} \right)^m$$

Problem: Trust Bank offers a savings account with a nominal discount rate of 4.8% payable monthly. Find the annual effective rate of interest i and the annual effective rate of discount d .

Solution.

4.8% discount monthly

$$1 - d = \left(1 - \frac{.048}{12} \right)^{12} \Rightarrow d = .0469579$$
$$i = .0492716$$

Fact: $i > i^{(m)} > d^{(m)} > d$ for all $m > 1$

Problem: Given that $d^{(4)} = 4.6\%$, find $i^{(12)}$.

Solution.

$$d = 1 - \left(1 - \frac{.046}{4}\right)^4 = .045212566$$

$$i = \frac{d}{1-d} = .0473535411133$$

$$i + 1 = \left(1 + \frac{i^{(12)}}{12}\right)^{12}$$

$$\Rightarrow i^{(12)} = .046355851$$

1.11 Constant Force of Interest

Suppose we have compound interest w/ annual effective rate i

Suppose $m_1 < m_2$ integers and $\underbrace{i^{(m_1)}, i^{(m_2)}}_{\text{nominal rates}}$

Note: $i^{(m_2)} < i^{(m_1)} < i$

So as m increases, $i^{(m)}$ decreases!

What happens when m increases to ∞

$$\frac{i^{(m)}}{m} = (1+i)^{\frac{1}{m}} - 1$$

$$i^{(m)} = m \left((1+i)^{\frac{1}{m}} - 1 \right)$$

$$\lim_{m \rightarrow \infty} m \left((1+i)^{\frac{1}{m}} - 1 \right)$$

$$= \lim_{m \rightarrow \infty} \frac{(1+i)^{\frac{1}{m}} - 1}{\frac{1}{m}} = \frac{0}{0}$$

By L'Hopital's Rule!

$$= \lim_{m \rightarrow \infty} \frac{(1+i)^{\frac{1}{m}} \cdot \ln(1+i) \cdot \frac{-1}{m^2}}{\frac{-1}{m^2}}$$

$$= \ln(1+i)$$

Definition. The force of interest is

$$\delta = \ln(1+i)$$

$$\begin{aligned} \hookrightarrow 1+i &= e^{\delta} \\ i &= e^{\delta} - 1 \end{aligned}$$

nominal
rate convertible
continuously!

so if $a(t) = (1+i)^t$
then $a(t) = e^{\delta t}$

Exercise: Show $\lim_{m \rightarrow \infty} d^{(m)} = \delta$ too!

$$i > i^{(m)} > \delta > d^{(m)} > d$$

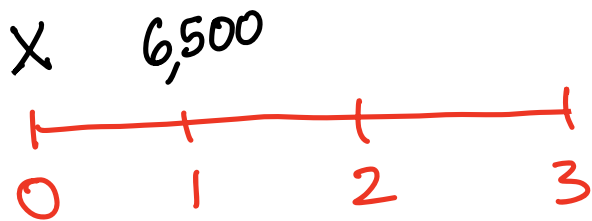
Problem: Estelle deposits \$12,500 in a four-year certificate of deposit at Community Trust Bank. If the annual percentage yield (APY) is 4.235%, find the nominal interest rate δ convertible continuously.

solution

$$i = .04235 \Rightarrow \delta = \ln(1.04235) \\ = .041477794463$$

Problem: Money at Swift National Bank grows at 3.8% compounded continuously. Rafael Ortiz closed his savings account at Swift exactly three years after he opened it, and he received his balance which was \$24,812. Mr. Ortiz made a \$6,500 deposit exactly one year after he opened the account, but he made no other deposits or withdrawals except for an unspecified opening deposit X . Find X and his balance exactly two years after he opened the account.

Solution. $\delta = .038 \Rightarrow i = e^{\delta} - 1$
 $= .03873123$



$$X(1+i)^3 + 6,500(1+i)^2 = 24,812$$

$$= X e^{.038(3)} + 6,500(e^{.038(2)}) = 24,812$$

$$X = 15,881.0702856$$

after 2 years

$$= X(1+i)^2 + 6500(1+i)$$

$$= X e^{2\delta} + 6500 e^{\delta} = 23,886.833489$$

1.12 Force of Interest

$$i = \frac{a(t+1) - a(t)}{a(t)}$$

tells you interest from t to $t+1$

but sometimes it doesn't give you a good indication of how the account is doing at time exactly t .

$$\frac{a(t + \frac{1}{m}) - a(t)}{a(t)}$$

will do better!
especially as $m \rightarrow \infty$

$$\begin{aligned} \hookrightarrow i^{(m)} &= \frac{\frac{a(t + \frac{1}{m}) - a(t)}{a(t)}}{\frac{1}{m}} = m \left(\frac{a(t + \frac{1}{m}) - a(t)}{a(t)} \right) \end{aligned}$$

$$\text{as } m \rightarrow \infty$$

$$\frac{1}{m} \rightarrow 0$$

$$m \left(\frac{a(t + \frac{1}{m}) - a(t)}{a(t)} \right)$$

$$= \frac{a'(t)}{a(t)} = \text{Force of interest } \delta_t \text{ at time } t.$$

Problem: Suppose that you have simple interest at a rate r . Find the force of interest δ_t .

Solution. $a(t) = 1 + rt \Rightarrow a'(t) = r$

$$\delta_t = \frac{r}{1 + rt}$$

Problem: Suppose that you have compound interest at a rate i . Find the force of interest δ_t as a function of i .

Solution. $a(t) = (1 + i)^t$

$$a'(t) = (1 + i)^t \ln(1 + i)$$

$$\delta_t = \ln(1 + i) = \delta$$

calculus fun fact:

$$\frac{d}{dx} \ln(u) = \frac{u'}{u} \quad \text{so...}$$

$$\delta_t = \frac{a'(t)}{a(t)} = \frac{d}{dt} \ln(a(t))$$

Problem: Suppose $a(t) = (1.05)^{\frac{t}{2}} (1.04)^{\frac{t^2}{3}} (1.03)^{\frac{t^3}{6}}$. Find δ_t .

Solution.

$$a(t) = (1.05)^{\frac{1}{2}t} \underbrace{\left[(1.04)^{\frac{t^2}{3}} (1.03)^{\frac{t^3}{6}} \right]}_g$$

$\downarrow$
 f

$$f' = \frac{1}{2} (1.05)^{\frac{1}{2}t} \ln(1.05)$$

$$g = \underbrace{(1.04)^{\frac{t^2}{3}}}_u \underbrace{(1.03)^{\frac{t^3}{6}}}_v$$

$$u' = \frac{2t}{3} (1.04)^{\frac{t^2}{3}} \ln(1.04)$$

$$v' = \frac{3t^2}{6} (1.03)^{\frac{t^3}{6}} \ln(1.03)$$

$$g' = \underbrace{\frac{1}{3}t^3 (1.04)^{t^2/3} (1.03)^{t^3/6}}_{u'} \underbrace{\ln(1.03)}_v$$

WAIT this is way too much work!

$$\frac{d}{dt} \ln(a(t)) = \frac{d}{dt} \left[\ln \left((1.05)^{\frac{t}{2}} \cdot (1.04)^{t^2/3} (1.03)^{t^3/6} \right) \right]$$

$$= \frac{d}{dt} \left[\frac{t}{2} \ln(1.05) + \frac{t^2}{3} \ln(1.04) + \frac{t^3}{6} \ln(1.03) \right]$$

$$= \frac{1}{2} \ln(1.05) + \frac{2t}{3} \ln(1.04) + \frac{3t^2}{6} \ln(1.03)$$

If $\delta_t = \frac{d}{dt}(\ln(a(t)))$

dummy variable!

$$\int_0^t \delta_t \, dr = \int_0^t \frac{d}{dr} (\ln(a(r))) \, dr$$

FTOC $\left\{ \begin{array}{l} = \ln(a(t)) - \ln(a(0)) \\ = \ln(a(t)) - \ln(1) \\ = \ln(a(t)) \end{array} \right.$

so

$$a(t) = e^{\int_0^t \delta_r \, dr}$$

Problem: Suppose $\delta_t = \frac{3}{1-3t}$. Find the corresponding accumulation function $a(t)$.

Solution

$$\int_0^t \frac{3}{1-3r} dr$$

$$u = 1-3r$$

$$du = -3 dr$$

$$-du = 3 dr$$

$$-\int_0^t \frac{du}{u} = -\ln(u) = -\ln(1-3r) \Big|_0^t$$

$$= -\ln(1-3t)$$

$$a(t) = e^{-\ln(1-3t)} = (1-3t)^{-1}$$